Impact Brands: one trusted data model for 7 brands

OVERVIEW

From public storefront data to a tested marketing & subscription warehouse (dbt · BigQuery-ready · Looker Studio)

PUBLIC · page 1

207

products across 7 storefronts, all sold through one offer logic

PUBLIC + SIMULATED · page 4

100%

of sales and ad spend mapped to a product after the fix at the source

SIMULATED · page 7

2.45 vs 1.71

platform ROAS vs net first-party ROAS: the gap, explained layer by layer

SIMULATED · page 8

9 of 10

injected data defects caught by dbt tests before a dashboard showed them

Business questions

1. **Portfolio** What do the 7 brands sell, and how alike are they? p.1-3
2. **Mapping** Can every SKU, campaign and ad account be tied to one product and brand? p.4
3. **Warehouse & metrics** What does a trusted model and metric layer look like? p.5-6
4. **Reconciliation** Why do Media Buying and Finance see different ROAS? p.7
5. **Quality & reporting** Do the tests catch problems before users do? p.8-9

Recommendation: keep product, brand and ad-account mapping in versioned seeds tested on every build, so a new product or account fails loudly on day one; report ROAS on first-party orders next to platform ROAS and MER; publish one definition of active subscription, active customer and churn, and build every dashboard on the same marts.

Data & methods

PUBLIC real catalogue of the 7 storefronts (products, bundles, subscribe & save prices, store SKU codes), archive history, Google Trends, company facts.

SIMULATED orders, subscriptions, payments, 3PL, Meta and Google Ads, CRM: generated on top of the real catalogue, with 13 problems injected on purpose. Not Impact Brands data.

dbt staging → intermediate → marts, versioned mapping seeds, generic + business-rule tests, alerts.

Stack DuckDB locally, same project on BigQuery; Looker Studio / Power BI on the reporting marts; Dataform version of 2 models.

How to read this report

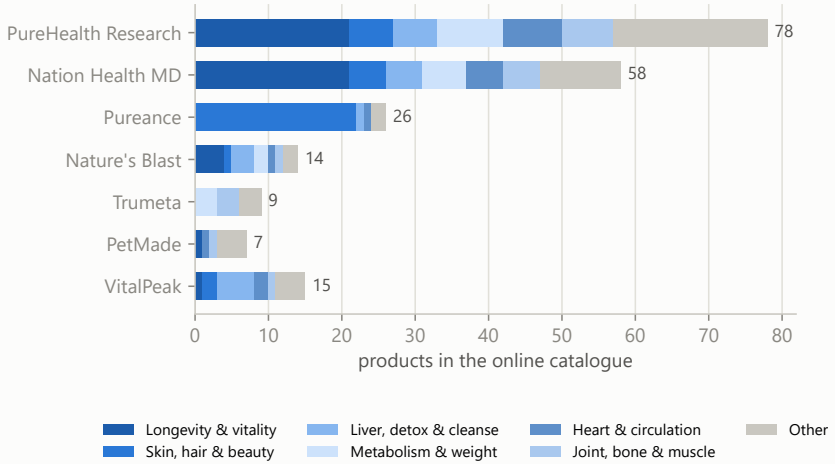
Titles state the conclusion. **Tags** PUBLIC DATA / SIMULATED / PUBLIC + SIMULATED mark where the data comes from. **Red** marks gaps and alerts, **dashed lines** thresholds. **Code**: every number is rebuilt by the scripts and the dbt project in the repository.

PureHealth Research and Nation Health MD carry 66% of the 207-product catalogue and 98% of on-site reviews

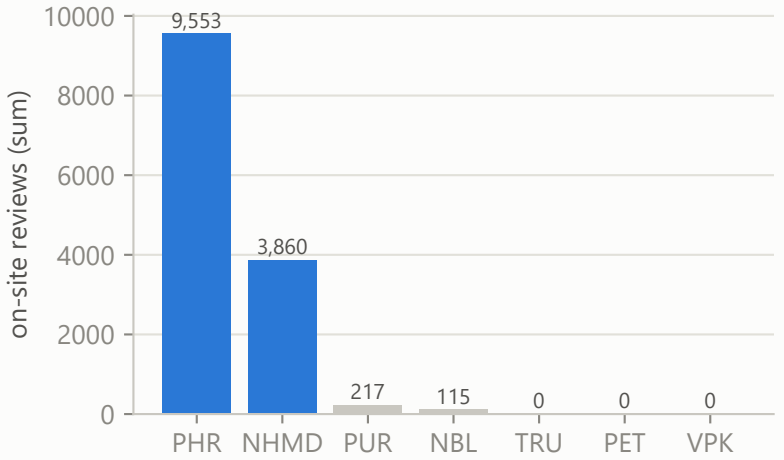
PUBLIC DATA

Seven brands share one platform and one offer logic. One canonical product and brand model can serve all of them; the two largest brands set the volume the data model must handle.

Catalogue size and mix by brand (category labels from product names)



Social proof is concentrated: on-site reviews by brand



Brand scorecard (snapshot 2026-11-11)

Brand	Focus	Products	Median price, 1 unit	Subscribe & save	On-site rating
PureHealth Research	Doctor-formulated supplements (flagship)	78	$59.99	100%	4.9
Nation Health MD	Doctor-formulated supplements	58	$49.50	100%	4.9
Pureance	Skincare and beauty supplements	26	$49.00	81%	5.0
Nature's Blast	Powdered supplements	14	$59.99	100%	4.8
Trumeta	Fitness and sports nutrition	9	$29.00	89%	5.0
PetMade	Pet wellness	7	$39.00	100%	5.0
VitalPeak	Single-ingredient and value supplements	15	not shown	-	-

The company in public numbers

402
team members (Aug 2025 press release)

3.3M
people reached, per the company

7
brands in the job post, 7 storefronts verified

1
shared stack: same storefront and checkout platform on 6 stores

Six stores show the same company address in their footer; VitalPeak is linked from the group site. Remote-first, data roles based in Lithuania. A "Tracking Systems & Data" team sits under the CTO.

Sources: 7 brand storefronts (sitemaps + product pages), PR Newswire 29 Aug 2025, theimpactbrands.com · collected 2026-11-11

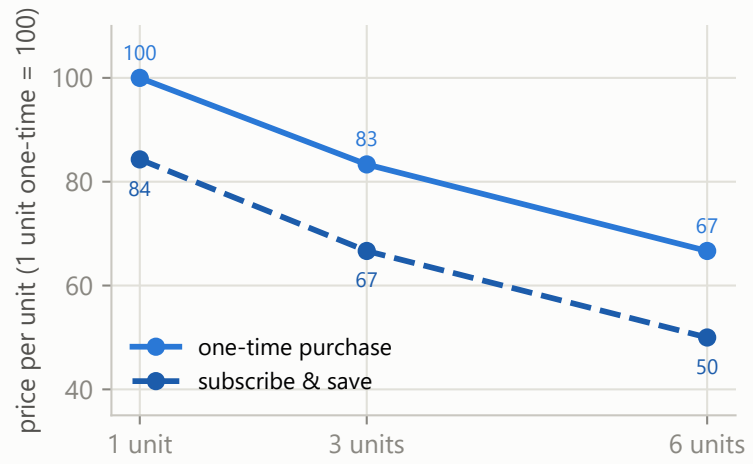
Independent case study based on public data · not affiliated with or endorsed by Impact Brands

Every brand sells the same ladder: 6 units cost 33% less per unit than 1, and 50% less on subscription

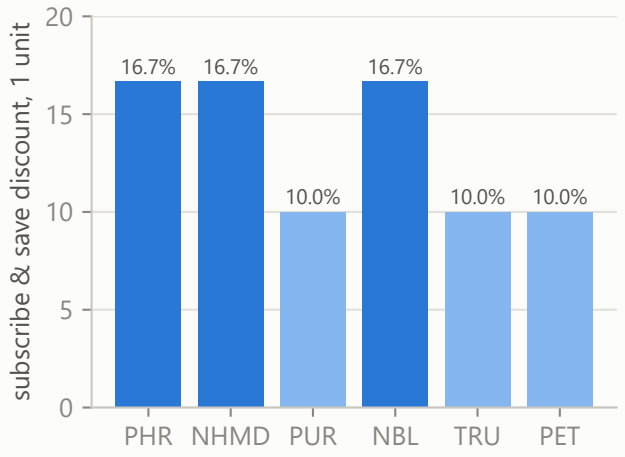
PUBLIC DATA

Prices, bundles and subscribe & save terms read from the live product pages; price history and launch pace from archived copies. One pricing logic across 6 stores is good news for a shared data model.

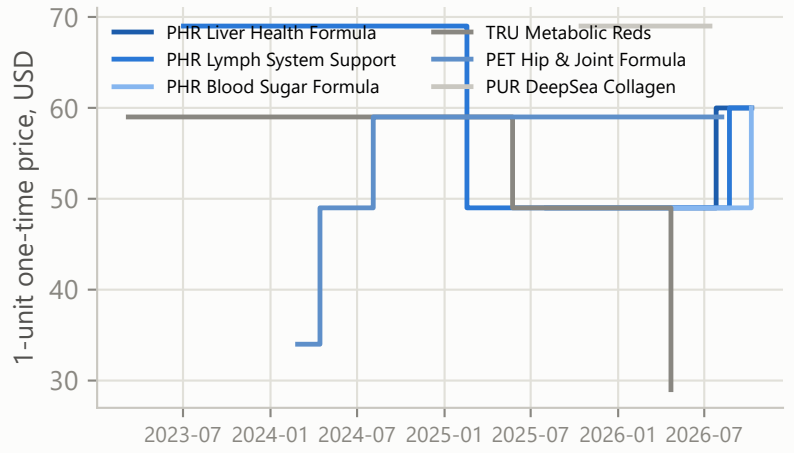
The bundle ladder: price per unit by bundle size (median of all products)



Two subscribe & save policies: 16.7% vs 10%



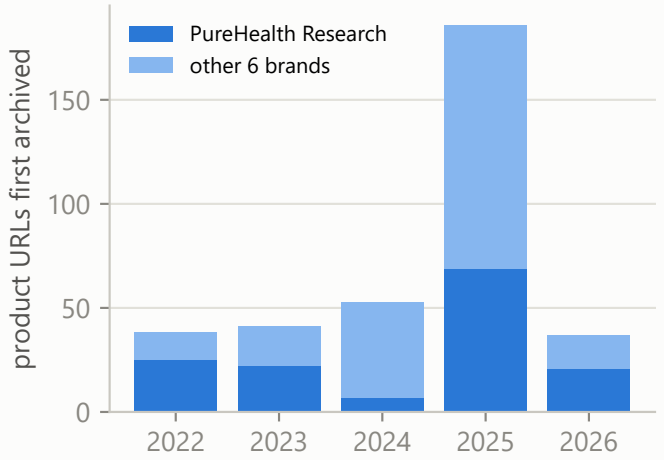
Archived 1-unit prices move in steps (Wayback Machine)



Why this matters for the data model

- One product is sold as **6+ sellable offers** (1/3/6 units x one-time / subscription), each with its own store code. Revenue per product needs a mapping from every code to one product and a units-per-item factor.
- A crossed-out list price sits 26% above the 1-unit price: "discount" must be defined once (vs list or vs 1-unit price).
- Subscriptions start at a discount and renew automatically: active subscriptions and churn need precise definitions (page 6).

Product page URLs first seen in the archive (new and renamed pages)



Sources: storefront product pages (prices per unit, crossed-out list price) · Internet Archive CDX, monthly captures since 2023 · collected 2026-11-11

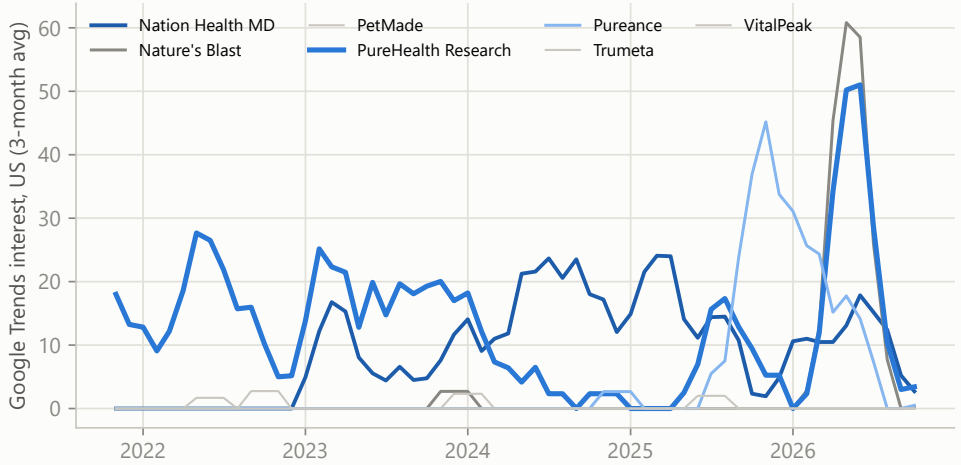
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Only 2 of 7 brands show up in branded search most weeks: demand is bought through paid media, so ad data has to be right

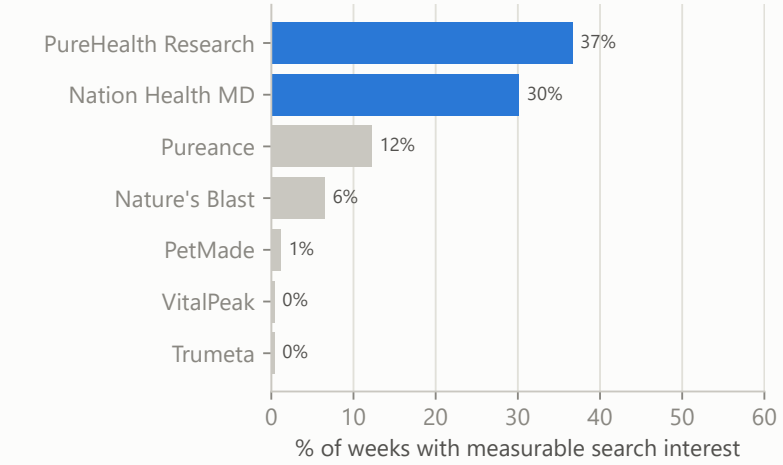
PUBLIC DATA

Google Trends, United States, weekly over 5 years. A signal of where demand comes from, with clear limits.

Branded search interest over 5 years (relative index 0-100, one shared scale)



Most brands rarely register in branded search



Reading this page

- **What it shows**: relative interest, not volumes. Two batches of terms were put on one scale through a shared anchor (PureHealth Research).
- **What it suggests**: demand for most brands is created by paid media, not found through branded search. The ad platform connectors, campaign naming and attribution are therefore the most business-critical pipelines.
- **Limits**: low-volume terms show as zero; "PetMade" and "VitalPeak" also match unrelated searches; ad libraries were not collected (login required).

Source: Google Trends (pytrends), US, 5 years, weekly, collected 2026-11-11. Relative index; not search volumes.

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Every sale and every ad dollar now maps to one of 207 products: 6.1% of recent sales were unmapped before the fix

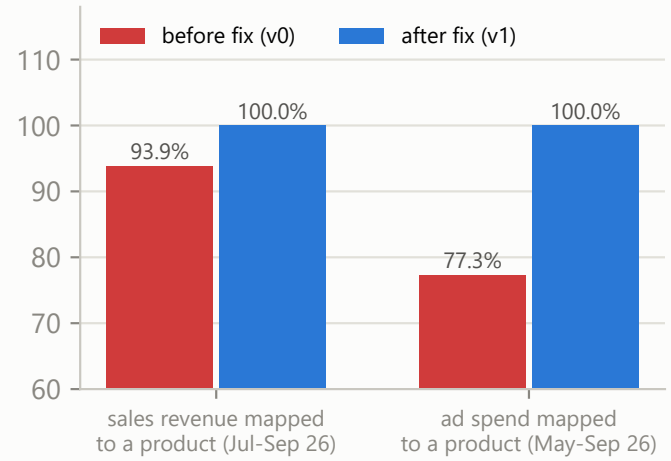
PUBLIC + SIMULATED

The public catalogue shows three SKU code styles, a store code that differs from the brand name and near-identical products across brands. Mapping rules are versioned seeds, tested on every build.

Why mapping is hard: real codes and names from the public stores

Brand	Product	One-time store code(s)	Subscription / other
PureHealth Research	Liver Health Formula	LHFU, quantity 1 / 3 / 6 (code per bottle)	LHFU1AD / 3AD / 6AD (code per bundle)
Nation Health MD	ActiveSpan	ATK, quantity 1 / 3 / 6	ATK3AD · store code is LHP, not NHMD
PetMade	Calm Soft Dog Chews	CFOF & QUANTITY=3 (third link style)	CFOF3AD, quantity 3
Trumeta	Bone Broth Protein	TBPB1 x 3 but TBPB6B for 6 (two styles)	TBPB3SS (SS suffix, not AD)
Same name, two brands	Bone Broth Protein	PHR: TXM	TRU: TBPB1 · two products, not one
Same need, four brands	Liver support	PHR Liver Health Formula · NHMD Liver Renew	NBL Liver 360 · VPK Liver Cleanse & Detox

Mapping rate before and after the fix



Before the fix, $161K of sales from 01 Jul 2026 had no product: the new product's SKUs were not in the mapping.

The rules live in versioned seeds, not in the report (seeds/mapping/*.csv)

Source	Source code	Canonical product / brand	Units	Fixed in v.
shop	LHFU3AD	PHR:liver-health-formula-supplement	3	0
warehouse	WH-PHR-10308	PHR:liver-health-formula-supplement	1	0
campaign rule	token 'bloodsugar'	PHR:blood-sugar-formula-supplement	-	0
campaign rule	token 'bloodsugr' (alias)	PHR:blood-sugar-formula-supplement	-	1
shop	YCH1AD (new product)	PHR:yeast-protein-creatine-electrolytes	1	1
ad account	act_...99777	VPK	-	1

Fix at the source, not in the chart

1. **Alert** `assert_no_unmapped_sales` fails on the first sale of SKU `YCH1AD`.
2. **Fix**: one pull request adds the new codes to `product_mapping.csv` (fixed_in_version = 1), reviewed like code.
3. **Prevent**: the same test runs on every build, so the next launch is caught on day one. Campaign typos get an alias *and* are renamed in Ads Manager.

758 real store codes and the 3PL codes map to 207 canonical products, key `brand_code:product_slug`.

Codes and products: public storefronts (2026-11-11). Sales, spend and mapping rates: SIMULATED (seed 42), before (fix_version 0) vs after (1).

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One dbt project, 33 models and 84 tests: runs locally on DuckDB, written to run unchanged on BigQuery

SIMULATED

Each layer has one job: staging cleans each source, intermediate holds the business logic (mapping, subscription periods, payment matching), marts serve each team. Cross-database macros keep it portable.

Sources → raw → staging → intermediate → marts → dashboards

SOURCES
- Store
- Subscriptions
- Payments
- 3PL
- Meta · Google Ads
- CRM · Finance

→

RAW
- 11 tables
  as received
- seeds
  9 CSV: public catalogue, mapping rules, FX

→

STAGING · 12
- 1 model per source
  rename, types, UTC, cents→units, dedupe

→

INTERMEDIATE · 7
- product & brand mapping
- subscription periods
- order ↔ payment match

→

MARTS · 14
- dims: brand, product, customer, date, ad account
- facts: orders, lines, payments, ad spend, subs daily
- mart_media_buying · mart_finance_reconciliation · mart_subscription_health · mart_roas_bridge

→

BI
- Looker Studio
- Power BI

dbt lineage graph (dbt docs, generated from the project)



Star schema and project in numbers

- dim_date
- dim_brand
- dim_product

fct_orders · fct_order_lines
fct_payments · fct_ad_spend_daily
fct_subscriptions_daily

- dim_customer
- dim_ad_account
- seeds: mapping

33
dbt models (staging 12, intermediate 7, marts 14)

84
tests: 72 generic + 12 business rules

2
targets: DuckDB (built and tested here) and BigQuery (parsed; needs a Google account to run)

0
engine-specific SQL outside 5 small dispatch macros

Counting the subscription tool's ACTIVE status overstates active subscriptions by 4.8%: paused and failing ones are not active

SIMULATED

On 30 Sep 2026: 11,565 naive vs 11,035 documented (449 paused, 81 past due). Average monthly churn 10.5%.

One definition per metric, written once

Metric	Definition (METRICS.md, with SQL)
active_subscription	state = active on that day: not paused, no failed payment being retried, not cancelled
active_customer	≥ 1 order with positive net revenue in the last 365 days, or an active subscription
churned_subscription	a cancelled event in the period; voluntary and involuntary split; reactivation counted apart
churn_rate	cancellations in month ÷ base at month start (active + paused + past due)
net_revenue	order total after discount − refunds − chargebacks known today, on the order month
CPA · ROAS · MER	spend ÷ first-party orders · first-party revenue ÷ spend · total net revenue ÷ total spend

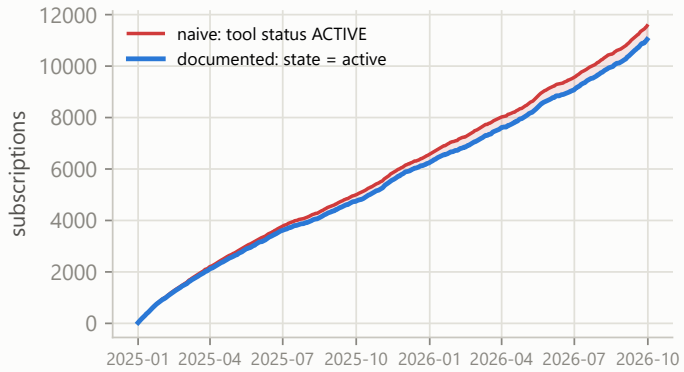
11,035
active subscriptions, 30 Sep 2026

44,987
active customers (365 days or subscribed)

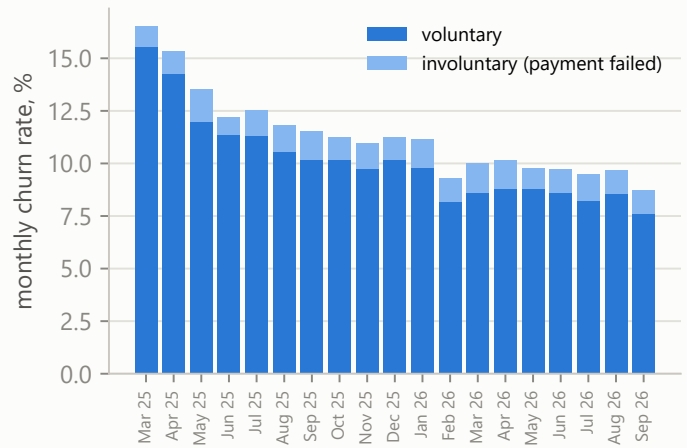
10.5%
average monthly churn, Mar 25 - Sep 26

11%
of cancellations are failed payments

Naive vs documented active subscriptions



Churn rate by month (cancellations ÷ base at start)



Edge cases decided, not left to each chart

- **Paused**: not active, stays in the churn base.
- **Payment failed**: "past due" while retries run; active again if a retry succeeds, otherwise involuntary churn (11% of all cancellations).
- **Reactivated**: active again from that day; its earlier cancellation still counts as churn.

The SQL behind the number

```
select date_day, count(*) as active_subscriptions
from marts.fct_subscriptions_daily
where is_active          -- state = 'active'
group by date_day
```

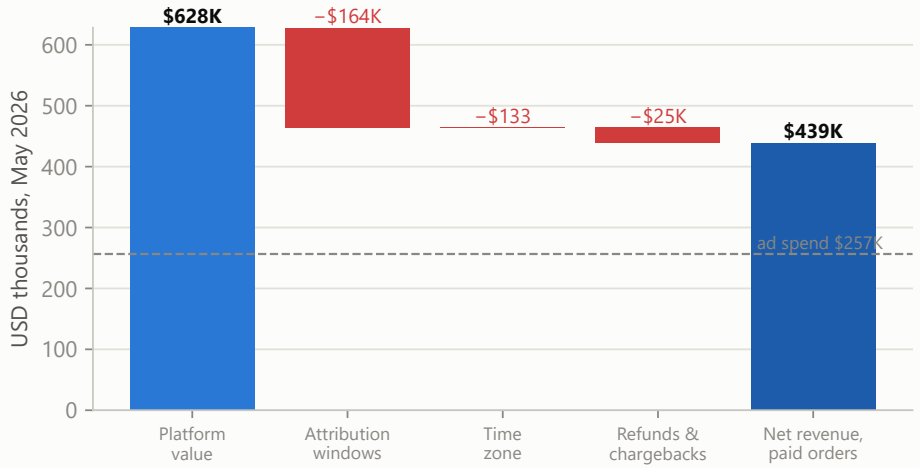

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Media Buying's ROAS of 2.45 and Finance's 1.71 are both right: attribution windows explain most of the gap

SIMULATED

An investigation from the ad platform down to the chart, with every difference quantified. The agreed numbers for May 2026: first-party ROAS 1.81 gross, 1.71 net, MER 2.87.

Revenue bridge for paid campaigns, May 2026 (mart_roas_bridge)



The question

Media Buying: **"ROAS is 2.45."**
Finance: **"ROAS is 1.71."**

Both are right about different numbers. Walking the number down each layer:

- **26%** of platform value is not backed by a real order (attribution windows).
- The time zone moves little over a month but **4%** of Google-attributed orders change day between LA and NY: it matters for daily pacing.
- Refunds and chargebacks take another **5.4%**.
- Before the fixes, spend was **$3K** short of the invoice (new account) and would have been **$7K** too high from the duplicate load.

Trace across every layer

Layer	What the number is	What changes it
1 · Ad platform UI	$628K value, 6,124 conversions	Platform counts view-through and 7-day clicks
2 · Raw connector tables	spend +$7K if read as is	14 May loaded twice (re-sync); new account missing 5 days
3 · Staging	spend $253K → $257K after backfill	dedupe latest load; invoice $257K matches
4 · Intermediate / marts	3,878 real paid orders, $464K gross	Orders joined on the campaign id in the UTM
5 · Chart (Finance)	$439K net	After refunds and chargebacks; NY business day

The tests caught 9 of 10 injected data defects before any dashboard showed them; the miss is documented, not hidden

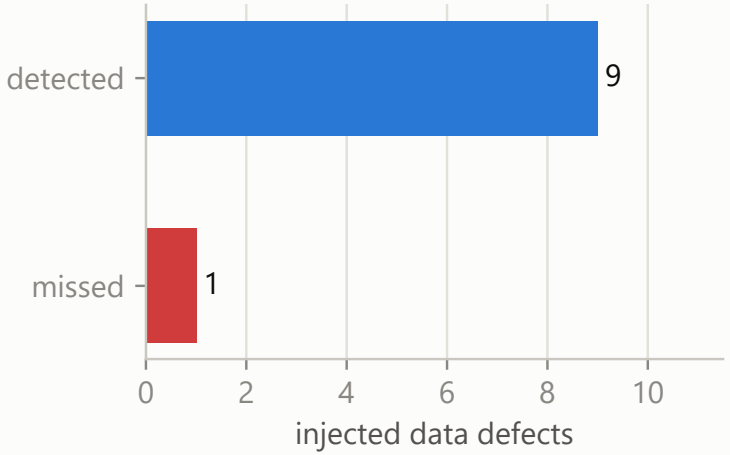
SIMULATED

Problems were injected on purpose and listed in a hidden answer key. The warehouse was replayed without the fixes (fix_version 0); a defect counts as caught only if its id appears in a failing test's stored rows.

Every injected problem and what caught it (answer key never read by the tests)

#	Injected issue	Outcome	Caught by
1	New ad account missing 5 days	detected	ad_spend_vs_platform_within_tolerance no_unmapped_ad_accounts
2	New product without mapping	detected	no_unmapped_sales
3	Typo in campaign name	detected	no_unmapped_campaigns
3	Wrong brand in campaign name	detected	campaign_brand_matches_account
3	Name matches the wrong product	missed	
4	Time zones (NY / LA / Vilnius / UTC)	explained (bridge)	
5	Webhook replays (485 orders)	detected	no_duplicate_order_events
5	Connector loaded a day twice	detected	no_duplicate_ad_rows
6	Dispute wave on a closed month	detected	closed_month_revenue_stable
7	Platform conversions > orders	explained (bridge)	
8	PayPal import in units, not cents	detected	orders_vs_payments_reconcile
8	EUR/GBP, cents vs units	explained (bridge)	
9	Paused/failing counted active	detected	active_subscriptions_exclude_paused_and_failed subscription_status_matches_events

Detection rate on data defects



The miss: a campaign named 'Sleep_StressRelief' matches the Sleep Formula rule and is silently attributed to it. Next test to add: flag campaign names that match more than one product rule.

The nightly alert (docs/alerts_example.md, from run_results.json)

> **Data quality alert - 7 failing · 5 warnings**
>
> - Active subscription count includes paused or failing ones: 6,736 row(s) · owner: Data
> - Sales on a store SKU with no product mapping: 4 row(s) · owner: Data
> - Campaign with spend that matches no product: 2 row(s) · owner: Media Buying
> - Warehouse spend differs from the platform invoice: 1 row(s) · owner: Data
> - Campaign name says another brand than its ad account: 1 row(s) · owner: Media Buying
> - Ad account sending spend without a brand: 1 row(s) · owner: Data / Media Buying
> - Warehouse item code with no product mapping: 1 row(s) · owner: Data / Ops
>
> + source freshness: raw_shop_orders STALE (extract ends 30 Sep)

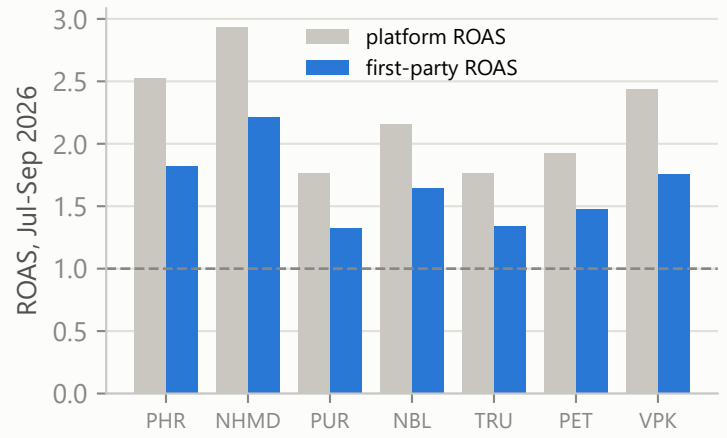
Three teams, three views, one set of definitions: every number on these pages comes from the same marts

SIMULATED

Mini versions of the Looker Studio pages (sources = the 3 reporting marts; ratios as calculated fields from summed columns). Same model works in Power BI.

Media Buying

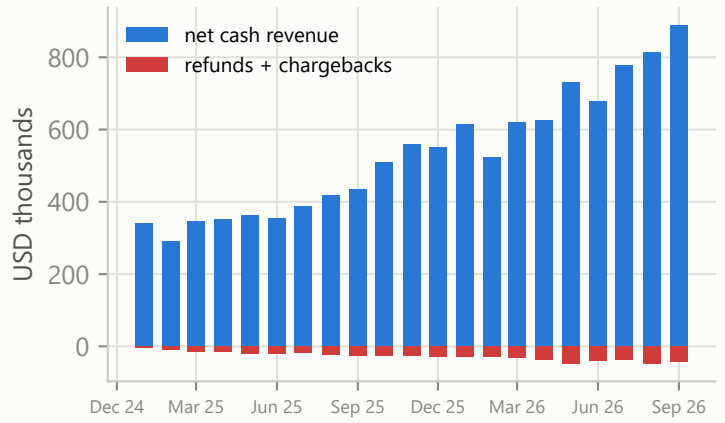
Platform vs first-party ROAS by brand



Brand	Spend	CPA	ROAS
PHR	$502K	$70	1.82
NHMD	$81K	$60	2.22
PUR	$67K	$78	1.33
NBL	$49K	$69	1.64
TRU	$37K	$65	1.34
PET	$33K	$66	1.48
VPK	$105K	$61	1.76

Finance

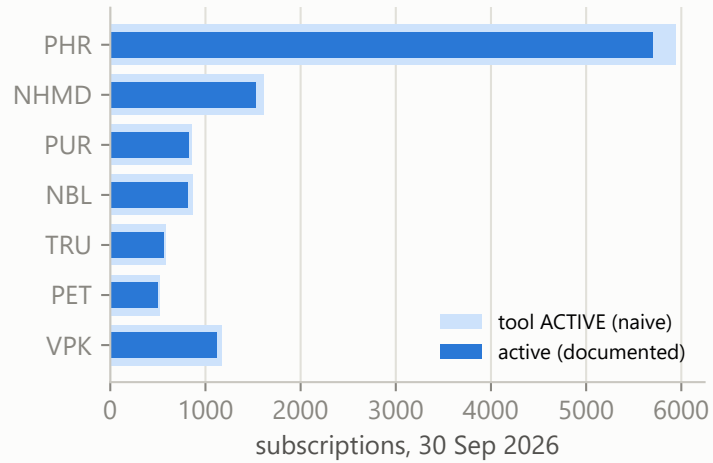
Net cash revenue and refunds by month



Month	Shop sales	PSP charges	Gap	On earlier months
Jul 2026	$818K	$818K	$0	$28K
Aug 2026	$863K	$863K	$0	$34K
Sep 2026	$934K	$934K	$0	$31K

Operations

Active subscriptions by brand



Brand	Active	Churn Sep
PHR	5,695	8.6%
NHMD	1,530	7.9%
PUR	820	8.7%
NBL	810	8.3%
TRU	562	9.5%
PET	495	8.8%
VPK	1,123	10.8%

Methods, limits and how to reproduce every number

Everything in this report is regenerated from code in the repository.

Data

- **PUBLIC**: 7 storefronts (sitemap → product pages, 1 snapshot, 2.5 s between requests, robots.txt respected), Wayback Machine CDX, Google Trends, company press release and careers page. Sources and dates: `data/SOURCES.md`.
- **SIMULATED**: 21 months of orders, subscriptions, payments, shipments, ad delivery, invoices and CRM contacts, built on the real catalogue (`src/simulate.py`, seed 42). Assumptions: `data/ASSUMPTIONS.md`.
- Skipped: Trustpilot (robots.txt disallows crawlers), ad libraries (login), company registry (no clear match).

Methods

- **dbt**: staging per source, intermediate business logic, marts per team; seeds for public catalogue and versioned mapping rules; 5 adapter-dispatch macros for DuckDB/BigQuery.
- **Tests**: generic (unique, not null, relationships, accepted values) + 12 business rules with stored failures; source freshness.
- **Detection**: warehouse replayed without fixes (`--vars '{fix_version: 0}'`), failing rows matched to a hidden answer key.
- **Report**: Python + matplotlib → HTML → PDF (headless Edge).

Limits

- Simulated data shows methods, not Impact Brands' performance, volumes or systems.
- VitalPeak shows no price online: assumed price ladder in the simulation.
- Archive first-capture dates approximate launches; Trends is a relative index.
- Categories come from product names: reporting labels, not health claims.
- Dataform models are written for comparison, not run here.

Reproduce

```
pip install -r requirements.txt
python src/collect_storefronts.py   # optional: new snapshot
python src/build_public.py
python src/simulate.py
cd dbt && cp profiles.example.yml profiles.yml
dbt build --profiles-dir .                    # with fixes
dbt run --profiles-dir . --target duckdb_before_fix --vars "{fix_version: 0}"
dbt test --profiles-dir . --target duckdb_before_fix   --vars "{fix_version: 0}" --target-path target_before_fix
cd .. && python src/evaluate_detection.py
python src/alerts.py --target-path dbt/target_before_fix   --db dbt/impact_brands_before_fix.duckdb
python src/build_report.py
```

AI note

AI-assisted workflow: I used Claude, mainly to write scripts faster. I drove the project: I defined the questions, chose and validated the data sources, made the modelling decisions and checked every result.